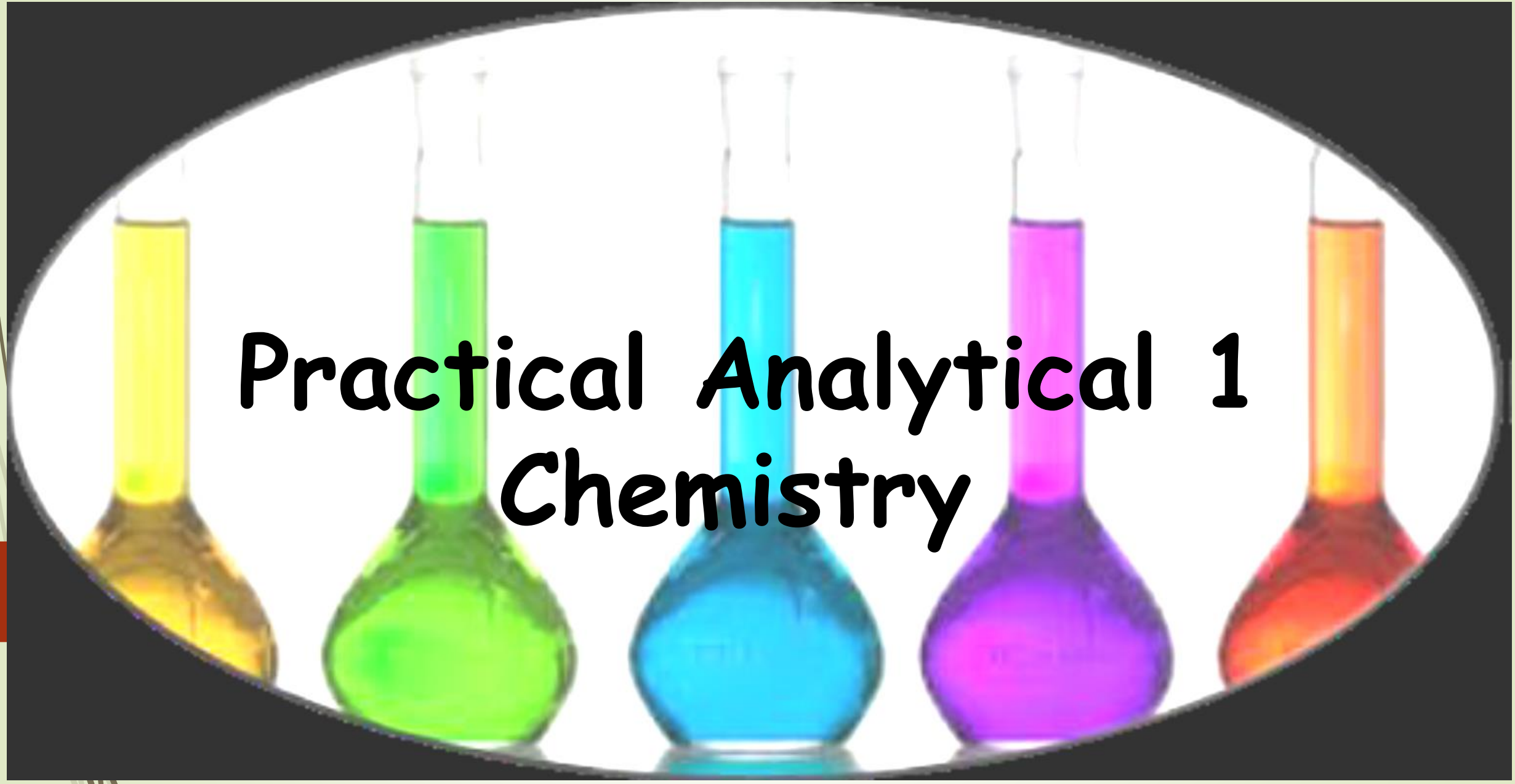


Practical Analytical 1 Chemistry

Experimental 4:

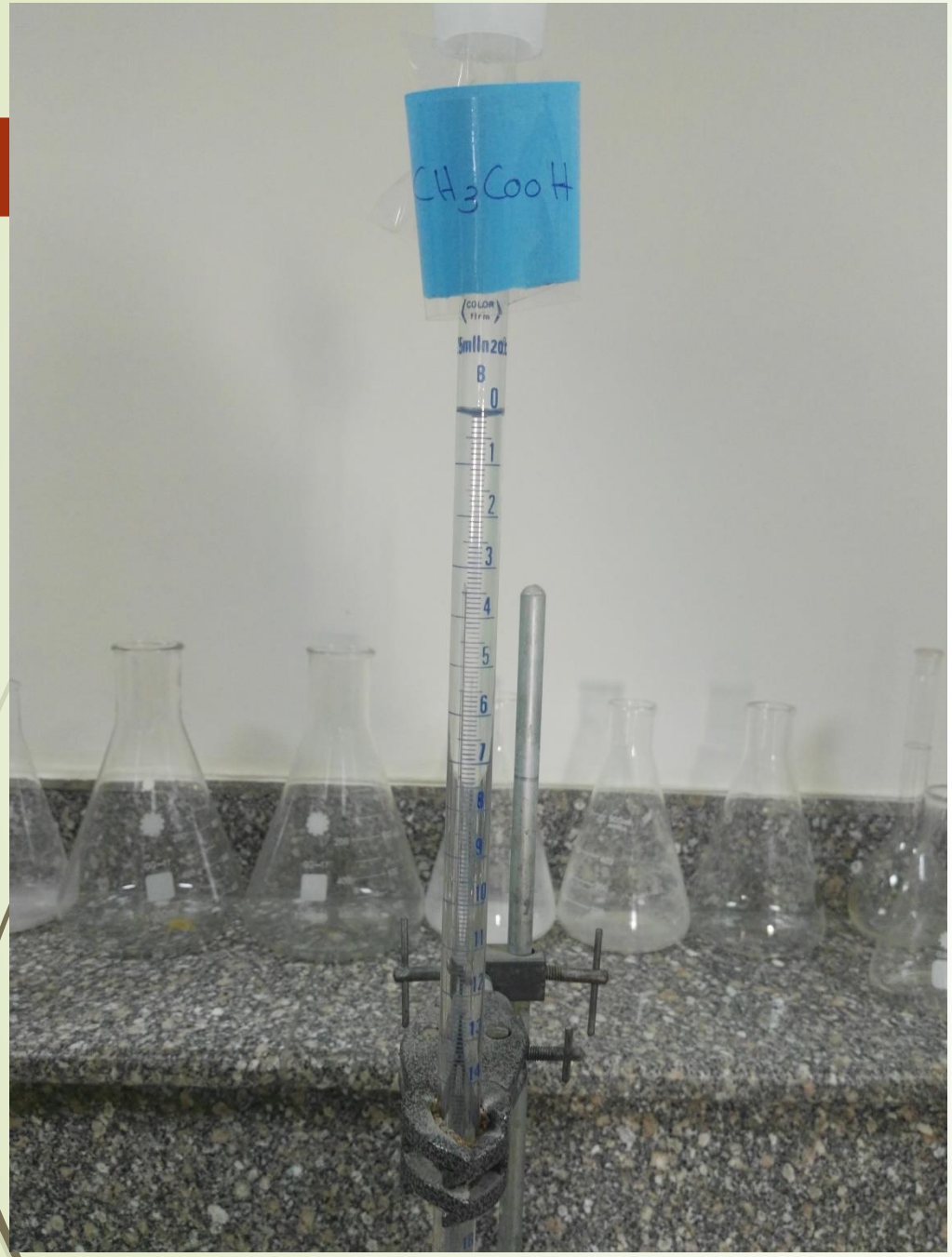
Standardization of CH_3COOH using standard solution of NaOH

Chemicals and tools:

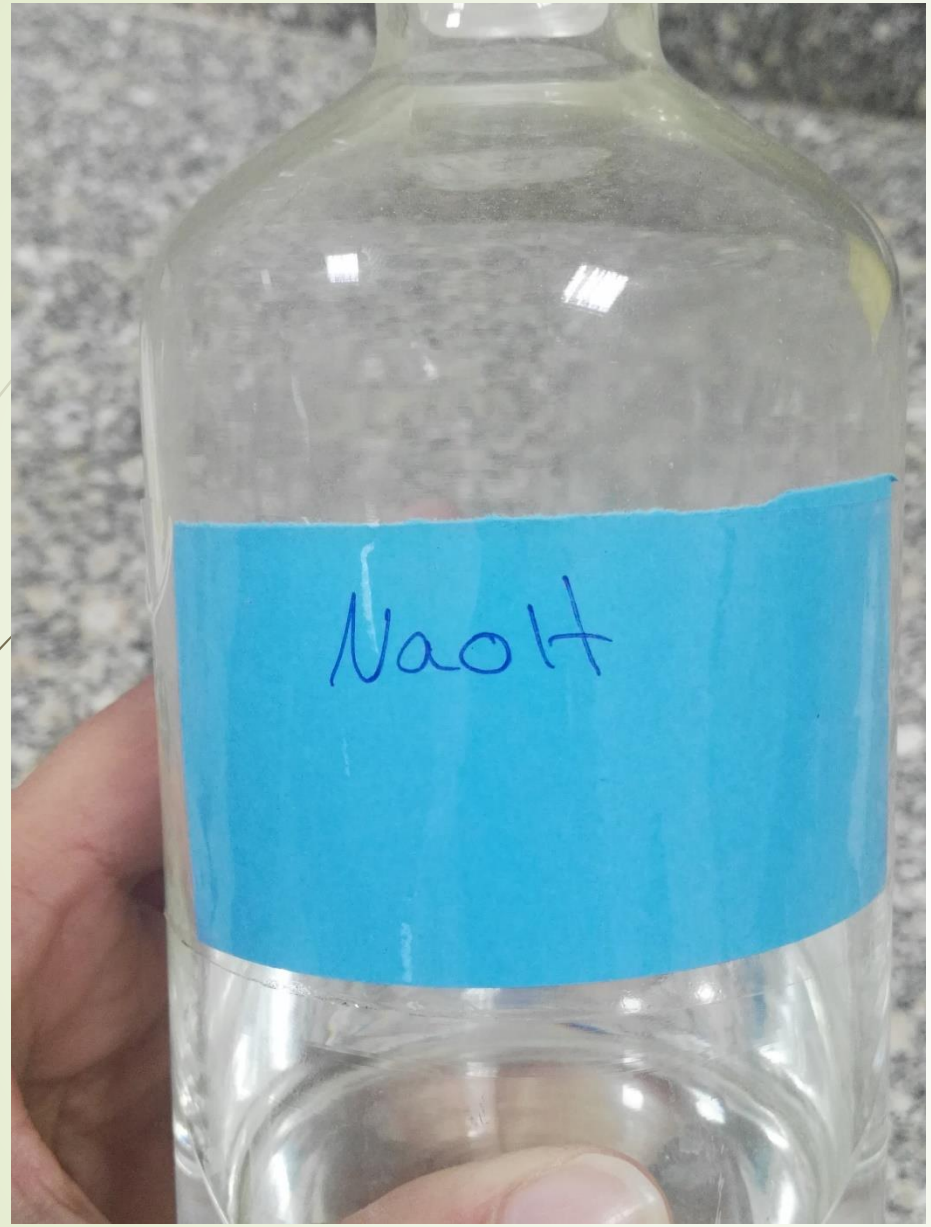
- 25-mL Burette
- 250-mL conical flasks
- Burette funnel
- 10-mL pipette
- Graduated Cylinder
- CH_3COOH(aq)
- NaOH(aq)
- Phenol Phethaline (ph.ph.)
- Distilled water

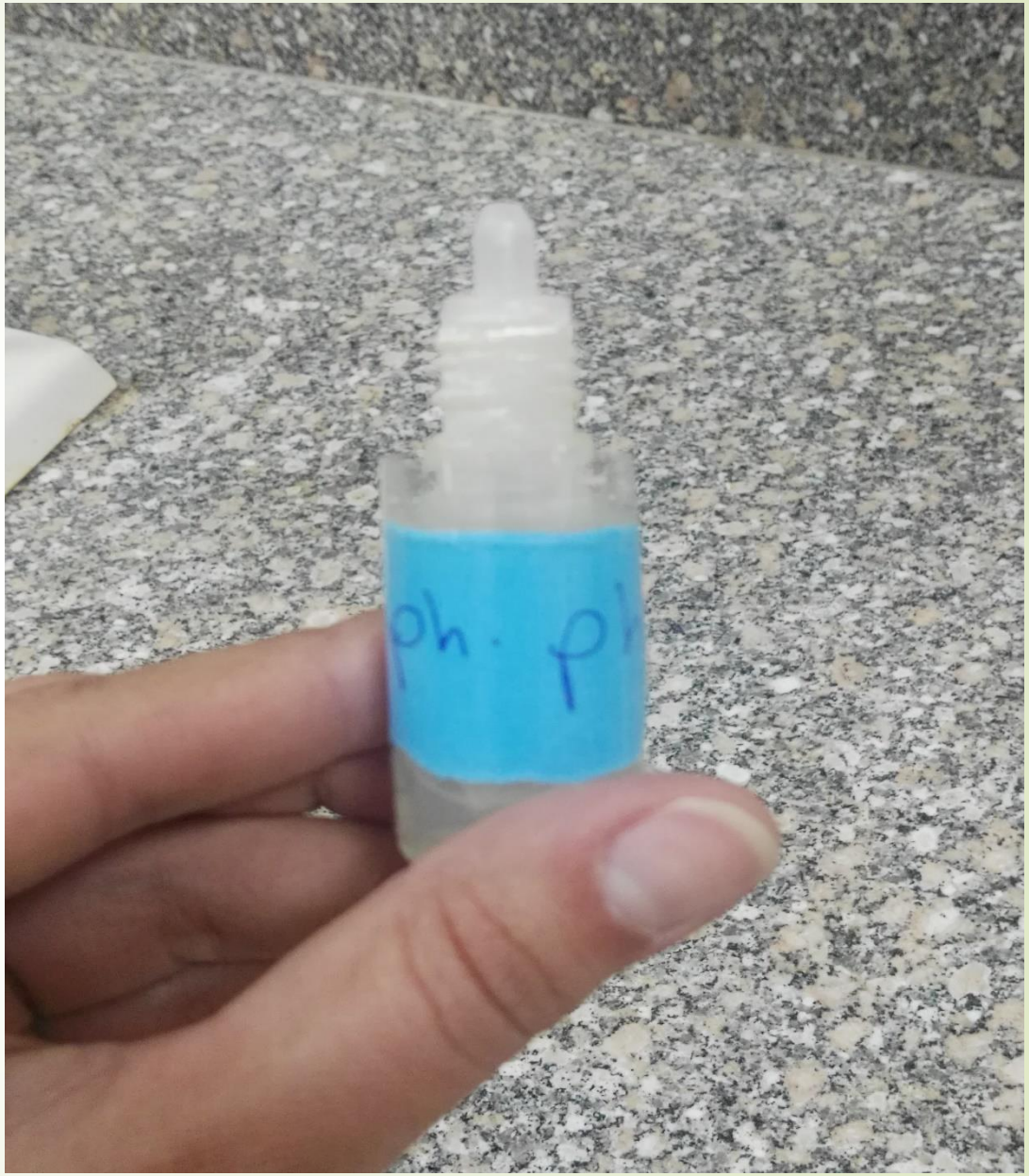
Procedure:

- Add the acid solution to the burette.
- Draw 10 mL of the base solution into the volumetric pipette and transfer this solution into a conical flask.
- Add 1-2 drops of ph.ph. to the base solution in the flask.
- Place the flask under the burette and start adding the acid solution to the conical flask, until the color change from purple to colorless.
- Record the final reading of the burette. Wash the contents of the flask down the drain with water.
- Calculate the concentration of NaOH.


CH3COOH

NaOH


NaOH


ph. pl

Calculation

- Suppose that the volume of CH_3COOH is V1 and its normality is N1 while V2 is the volume taken from NaOH and N2 is its normality.
- The volume of CH_3COOH (from burette) = V1
- $(N1 \times V1)_{CH3COOH} = (N2 \times V2)_{NaOH}$
- $(0.1 \times V1_{exp})_{CH3COOH} = (N2 \times 10)_{NaOH}$
- $0.1 \times 4 = N2_{NaOH} \times 10$
- $N2_{NaOH} = 0.04$ N
- Concentration in gm/L = normality x eq. wt.
- **Concentration of NaOH = N1 x 40= 0.04 x 40= 1.6 g/L**